

Gul e Fatima Kiani

ORCID: [0009-0007-8531-9975](https://orcid.org/0009-0007-8531-9975) |  [Google Scholar](https://scholar.google.com/citations?user=0009-0007-8531-9975)

Email: gkiani@unomaha.edu

Personal Website: gulefatimakiani.github.io

School of Interdisciplinary Informatics (Si2)

University of Nebraska at Omaha

Omaha, NE 68182

Education

- Ph.D. in Computing & Information Science**, University of Nebraska at Omaha, USA. 2023–on
- Dissertation: *Understanding Cross-Language Programming Knowledge Transfer as a Regulated Learning Process*
 - Advisor: [Dr. Brian Dorn](#)
- M.S. in Computer Science**, Pakistan Institute of Engineering and Applied Sciences, Pakistan. 2016–2019
- Thesis: *Rights Management System using Blockchain*
- M.Sc. in Information Technology**, Quaid-i-Azam University, Pakistan. 2013–2015
- Thesis: *Advanced Study on Scalable Energy-efficient Asynchronous Dissemination (SEAD) Routing Protocol for Wireless Sensor Networks*

Research and Teaching Experience

- Research Assistant**, BRIDGE Lab, University of Nebraska at Omaha, Omaha, USA 2024–on
- Mapping transfer pathways and course equivalencies between Nebraska community colleges/four-year institutions and receiving universities as part of the CS FUTURES project.
 - Evaluated an informed self-placement intervention for introductory programming and conducted student focus groups to examine students' experiences with the intervention.
 - Analyzed and visualized curricular pathways across UNO computing programs using the *Curricular Analytics* tool to examine prerequisite structures.
- Research Assistant**, MALICE Lab, University of Nebraska at Omaha, Omaha, USA 2023–2024
- Investigated whether generative AI models could identify phishing cues previously identified through crowd-sourced research.
- Program Leader & Lecturer**, Beaconhouse International College, Islamabad, Pakistan 2021–2023
- Worked as a franchise Program Leader & Lecturer for University of London and Liverpool John Moores University for the BS Computer Science and BS Software Engineering programs.
 - Mentored students on semester-long projects, providing guidance on project development and technical problem-solving.
 - Courses taught: Introduction to Programming, Web Development, Data Programming in Python, Network Security, and Theory of Automata.
- Visiting Lecturer**, International Islamic University, Islamabad, Pakistan 2019–2019
- Course taught: Introduction to Computing.

Publications

- [1] **G. F. Kiani** and B. Dorn, "Curiosity or Brittleness? A Situated View of Programming Knowledge Transfer Across Languages," accepted at *ACM Technical Symposium on Computer Science Education (SIGCSE TS)*, 2027. [To appear]
- [2] **G. F. Kiani** and B. Dorn, "Evaluating Informed Self-Placement in Introductory Programming Courses," accepted at *SIGCSE Virtual*, 2026. [To appear]
- [3] **G. F. Kiani**, [Understanding Cross-Language Programming Knowledge Transfer as a Regulated Learning Process](#), In *Proceedings of the 31st ACM Conference on Innovation and Technology in Computer Science Education V. 2 (ITiCSE 2026)*, Madrid, Spain, 2026.
- [4] B. Parkinson, **G. F. Kiani**, M. I. Frecker, and C. Toh, [Investigating Social Relevance and Knowledge Construction in Engineering Design Research Published in the Journals of Mechanical Design and Mechanisms and Robotics](#), *Journal of Mechanical Design*, 2026.
- [5] **G. F. Kiani**, S. Kang, A. Grover, and C. Toh, [Ethics of Justice and Ethics of Care: Using Generative AI with Speculative Fiction to Assess Students' Ethical Reasoning Skills](#), *ASEE Annual Conference & Exposition*, Charlotte, North Carolina, 2026.

- [6] B. Parkinson, **G. F. Kiani**, M. I. Frecker, and C. Toh, [Know Thyself: Investigating Social Relevance and Knowledge Construction in Engineering Design Research](#), *International Design Engineering Technical Conferences and Computers and Information in Engineering Conference*, Anaheim, California, 2025.
- [7] B. Parkinson, **G. F. Kiani**, M. I. Frecker, and C. Toh, [A Framework for Analyzing Goal Alignment and Social Relevance of Research Papers to Identify Impact of Women in Design Research Communities](#), *International Conference on Engineering Design (ICED)*, 2025.
- [8] **G. F. Kiani** and T. Kayani, [Real-time Violence Detection using Deep Learning Techniques](#), *3rd IEEE International Conference on Innovations in Computer Science & Software Engineering (ICONICS)*, 2022.

Posters

- [1] **G. F. Kiani**, Understanding Cross-Language Programming Knowledge Transfer as a Regulated Learning Process, *31st ACM Conference on Innovation and Technology in Computer Science Education V. 2 (ITiCSE 2026)*, Madrid, Spain, 2026.
- [2] **G. F. Kiani**, Bridging the Experience Gap: An Informed Self-Placement Approach to Introductory Programming, *57th ACM Technical Symposium on Computer Science Education V. 2 (SIGCSE 2026)*, St. Louis, MO, 2026.
- [3] **G. F. Kiani** and C. Toh, From witness to voice: A platform for sharing visitor reflections inspired by the art of Samuel Bak, *18th Annual Student Research and Creative Activity Fair*, Omaha, USA, 2025.

Conference Presentations

- [1] B. Parkinson and **G. F. Kiani**, "Know Thyself: Investigating Social Relevance and Knowledge Construction in Engineering Design Research", *ASME 2025 International Design Engineering Technical Conferences (IDETC)*, 2025. **[Nominated for Best Paper Award]**
- [2] **G. F. Kiani** and T. Kayani, "Real-time Violence Detection using Deep Learning Techniques", *3rd IEEE International Conference on Innovations in Computer Science & Software Engineering (ICONICS)*, 2022.

Awards and Honors

ACM Travel Grant , ITiCSE \$500	2026
Graduate Research and Creative Activity (GRACA) Grant , University of Nebraska at Omaha \$5,000	2025
Student Conference Travel Fund , ORCA, University of Nebraska at Omaha \$500	2025
Travel Grant , College of IS&T, University of Nebraska at Omaha \$500	2025
DED Special Initiative Grant , ASME IDETC 2025 \$1,000	2025

Skills

Research Methods: • Mixed-methods research • Statistical analysis • Interviews • Focus groups • Thematic analysis • Inductive and deductive coding • Survey research • User studies

Programming & Data Analysis: Python • C/C++ • R • Power BI

Databases: MySQL • SQLite

Misc.: Academic research, teaching, training, $\LaTeX$ typesetting and publishing

Service

Artifact Evaluation Committee, The 32nd Symposium on Operating Systems Principles (SOSP 2026)

External Reviewer (PC Member), Since 2025 ACM SIGCSE Technical Symposium

External Reviewer (PC Member), 2026 ACM SIGCSE Virtual

External Reviewer, 2026 HICSS

Student Volunteer, 2026 ACM SIGCSE Virtual, 2025 ASME IDETC

Organizing Committee Member, 2024 Nebraska GenCyber Camp for Middle school students

Instructional Materials Editor, 2024 Nebraska GenCyber Camp for Middle school students [\[GitHub Repository\]](#)

References

1. **Brian Dorn**, Assoc. Dean for Academic & Faculty Affairs, University of Nebraska at Omaha bdorn@unomaha.edu
2. **Michelle Friend**, Associate Professor, University of Nebraska at Omaha mefriend@unomaha.edu
3. **Andrea Grover**, Associate Professor, University of Nebraska at Omaha andragrover@unomaha.edu
4. **Christine Toh**, Assistant Professor, James Madison University tohcx@jmu.edu